A Multitree Genetic Programming-Based Feature Construction Approach to Crop Classification Using Hyperspectral Images

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***Abstract*— Feature construction has shown promise in improving the accuracy of crop classification by constructing high-level features. However, current feature construction methods often rely on domain knowledge and have a limited interpretability of the solutions. To address this, this study proposes a new genetic programming (GP) approach to automatically evolve solutions with high interpretability that can construct high-level features for crop classification from hyperspectral images. A flexible representation of multiple trees is proposed in the proposed GP approach to construct various types of high-level features from the original ones, simultaneously. To improve the search ability, a new offspring generation method is developed to dynamically guide the evolution of the population while improving the diversity of the population. The new approach wraps with three classification algorithms, i.e., support vector machine (SVM), naive Bayes (NB), and k-nearest neighbor (KNN), for crop classification on three datasets with different difficulties and tasks. The results demonstrate that the features constructed by the new approach can effectively distinguish different crop categories. The new approach achieves better performance than the compared GP-based method, classic methods, and deep learning methods in crop classification using hyperspectral images. Importantly, the proposed approach shows the high interpretability of the constructed features.**

***Index Terms*— Crop classification, feature construction, genetic programming (GP).**

Manuscript received 30 January 2024; revised 26 May 2024; accepted 3 June 2024. Date of publication 24 June 2024; date of current version 24 July 2024. This work was supported in part by the National Natural Science Foundation of China under Grant U23A20340, Grant 62376253, and Grant 62106230; in part by China Postdoctoral Science Foundation under Grant 2023M743185; in part by the Key Laboratory of Big Data Intelligent Computing, Chongqing University of Posts and Telecommunications Open Foundation, under Grant BDIC-2023-A-007; in part by the Marsden Fund of New Zealand Government under Grant VUW2115; in part by the National Science Challenge-Science for Technological Innovation Challenge (SfTI) under Grant 2019-S7-CRS; in part by New Zealand Ministry of Business, Innovation and Employment under Grant C09X1923; in part by New Zealand the Ministry of Business, Innovation and Employment (MBIE) Data Science Strategic Science Investment Fund (SSIF) under Grant RTVU1914; and in part by New Zealand MBIE Endeavour Research Programme under Grant UoCX2104. *(Corresponding author: Ying Bi.)*

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Digital Object Identifier 10.1109/TGRS.2024.3415773

I. Introduction

In order to address the increased demand for food caused by the growth in the global population, close monitoring of agricultural activities is crucial [1]. Crop classification and mapping is a fundamental task in the field of agricultural activities monitoring through the utilization of remote sensing techniques [2]. Crop classification serves the purpose of identifying different types of crops, enabling better management of cultivation, fertilization, pest, and disease control, and ultimately improving crop yield and quality. The use of hyperspectral imaging technology for crop classification has been widely recognized and can be effectively applied in the field of precision agriculture [3]. However, it is difficult to achieve accurate crop classification using hyperspectral images due to challenges, such as high dimensionality, noise, and high variations.

Different methods have been proposed for crop classification. Spectral indices are derived through the calculation and analysis of the relationships between the spectral reflectance of different bands. Kobayashi et al. [4] calculated 91 published spectral indices and demonstrated that the use of spectral indices can effectively improve crop classification accuracy. Phenological indicators record the changes within specific lifecycles of crops and can characterize the differences between crops. Bargiel [5] proposed a new method for identifying specific crop sequences by using crop phenological changes. Kussul et al. [6] developed a multilevel deep learning method for land cover and crop classification. Zhong et al. [7] proposed a remote sensing time-series classification framework based on deep learning for classifying summer crops using Landsat enhanced vegetation index (EVI) time series.

The abovementioned methods can effectively classify crops, but they have their limitations. Spectral indices and phe-

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nological features are low-level features extracted from raw hyperspectral images, which may ignore useful information. For example, the normalized difference vegetation index (NDVI) can reflect the overall growth of crops, but it has limited information about vegetation types and soil quality. Moreover, phenological features are influenced by climate and geographical location, which may lead to the loss of classification accuracy. Deep learning is a method that automatically learns high-level features from raw data, but its performance often depends on a large number of high-quality training samples [8]. However, collecting and processing a large number of labeled samples is difficult and time-consuming in remote sensing. Moreover, deep learning models are often complex and lack interpretability [9]. In land resource management, highly interpretable models can help decision-makers better understand the changing trends of vegetation types in farmland. Therefore, it is necessary to develop new methods for crop classification.

Feature construction can combine original features to construct new discriminative high-level features to improve crop classification performance. Feature construction is similar to feature selection since both are dimensionality reduction methods. However, unlike feature selection, which selects a subset of features from the current feature set, feature construction can generate new high-level features. For crop classification using hyperspectral images, feature construction can be used to reduce the dimensionality of the data and improve the classification performance. By constructing a small number of high-level features, the interpretability of the classification system may be improved. However, this has not been comprehensively investigated.

Due to its powerful search and learning ability, evolutionary computation (EC) has been successfully applied in various fields [10], [11], [12]. Genetic programming (GP) is an EC technique in which the populations are genetically bred based on the Darwinian principles of survival of the fittest. Compared to other EC algorithms, GP typically uses a flexible tree-based variable-length representation that has potentially good interpretability [13]. The tree-based representation enables GP a suitable method for feature construction. In addition, GP has powerful search ability and implicit feature selection, which allows it to handle large complex feature spaces in many difficult tasks, such as hyperspectral image classification.

However, existing GP-based methods for crop classification have certain limitations. Existing GP methods usually only utilize the spectral band information from hyperspectral images [14]. The process of feature extraction is from a relatively narrow aspect and fails to fully utilize existing general domain knowledge to extract more distinctive features of different types. This often leads to poor performance of GP on complex tasks. Additionally, existing works focus on improving the performance of GP by mainly developing new tree/program representations [15], [16], while research on GP search strategy is relatively rare. Lack of effective search strategy, including offspring generation, in GP, may fail to search for optimal solutions during the evolutionary process or lead to premature convergence and getting trapped in local optima.

To address the above limitations, this article proposes a new feature construction approach based on GP to effectively construct new high-level features to classify crops. It is a multitree GP approach with an adaptive offspring generation method, abbreviated as AMTGP. In addition to the original spectral band features of hyperspectral images, AMTGP also uses texture features and spectral indices features as inputs or the original features for GP to construct high-level features. AMTGP adopts a flexible multitree representation and a new offspring generation method, which encourages the generation of better solutions and ensures the diversity of the population during the evolutionary process. Deep learning methods can discover end-to-end data representations for specific problems, but they often require a large number of training data and have poor interpretability of the solutions [17]. In contrast, the proposed AMTGP approach can automatically search for solutions without domain knowledge and with potentially high interpretability. The new approach will be evaluated on three commonly used remote sensing datasets (i.e., Salinas, Longkou, and Honghu) of varying difficulty and compared with the state-of-the-art GP methods and non-GP methods. Further analysis will be conducted to gain a deeper understanding of the new approach.

The main contributions of this article are summarized as follows.

1) A GP-based feature construction approach with a multitree program structure (AMTGP) is proposed to automatically construct high-level features for crop classification. The proposed approach extracts three types of features as inputs of GP and constructs high-level features using AMTGP with a multitree program structure and an offspring generation method for crop classification. The proposed approach is more effective than most existing GP and non-GP methods on the Salinas, Longkou, and Honghu datasets.
2) The proposed approach includes a new feature extraction stage, which extracts spectral indices features and texture features after preprocessing the hyperspectral images. A multitree representation for AMTGP is proposed to better utilize these features together with the original bands for feature construction. To preserve feature interaction within the same type, the multitree representation of the AMTGP individuals allows each tree to evolve only one type of feature to ensure its specialization. Each tree can construct a flexible number of features by increasing the depth of the tree.
3) A new offspring generation method is employed to generate better promising individuals that cover the wide range of search space. Aside from some individuals being copied to the next generation, the remaining individuals in the population are divided into two groups. The first group generates individuals using dynamic crossover and mutation operators, encouraging mutual learning between the better individuals and mutation of the poorer individuals. The second group evolves the trees of individuals independently, ensuring randomness and diversity of the population. The results confirm the effectiveness of the new population generation method.

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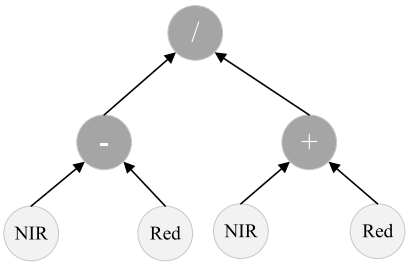


Fig. 1. Example tree.

Ensure the randomness in the composition of individuals of the other group to enhance diversity.

II. Background and Related Work

A. Genetic Programming

GP is an EC algorithm that can automatically evolve computer programs to solve a problem [18]. In GP, the computer programs can be represented in the form of trees, which is the most popular representation. A GP tree typically consists of root, internal, and leaf nodes. The root and internal nodes are formed using a predefined function set. The leaf nodes are formed using a predefined terminal set. The function set usually includes arithmetic and logical operators ($+$, $-$, $\times$, and protected $/$), as well as custom functions. The terminal set typically includes random constants and problem-specific features. Taking the vegetation index NDVI as an example, a tree is shown in Fig. 1. This example tree can be expressed as ProD(Sub(NIR, Red), Add(NIR, Red)), where ProD denotes the protected division. In this example tree, NIR is the near-infrared band and Red is the red band. These are combined to form NDVI.

In recent years, many GP-based algorithms have been proposed for image classification tasks. Bi et al. [19] proposed a two-stage-based GP method to evolve building blocks for feature extraction and evolve ensembles of classifiers, achieving good performance. Fan et al. [20] developed a surrogate-assisted GP method that uses global and local surrogate models to significantly reduce the computational cost while maintaining classification accuracy. To reduce the risk of overfitting, Bi et al. [21] first developed a GP method to solve a multitask feature learning problem as a multiobjective optimization problem and it improved the generalization. In [22], a multiobjective GP method was proposed, which evolved effective image descriptors by optimizing two objectives, i.e., the accuracy and the distance measure.

B. GP for Crop Classification

To address the problem of low efficiency in manual feature extraction, Lu et al. [14] proposed a GP-based feature learning method that utilized a single-tree representation to classify crops in time-series images. This representation enables GP to dynamically generate a fixed or flexible number of features without requiring domain knowledge. Wen et al. [23] proposed an object-based GP method for cropland field extraction in a binary classification task. This method constructed a GP classifier by segmenting the image, effectively extracting cropland. Batista et al. [24] proposed multidimensional multiclass GP (M3GP) (a variant of the standard GP algorithm) to improve the classification of land cover types by constructing hyper-features from satellite bands. dos Santos et al. [25] proposed an interactive technique for remote sensing image classification. The technique combines image region descriptors, allowing users to guide GP in learning user preferences by indicating areas of interest. Santos et al. [26] developed a GP-based method to combine texture and spectral information encoded by image descriptors for the automatic recognition of coffee crops in remote sensing images.

C. Other Methods for Crop Classification

Li et al. [27] proposed a novel method that combines the generative adversarial network (GAN), convolutional neural networks (CNNs), and long short-term memory neural network (LSTM) models for the classification of corn and soybean crops. Ji et al. [28] proposed a new method based on 3-D CNN for crop classification, outperforming three traditional methods and 2-D CNN. Mazzia et al. [29] proposed a novel and optimal deep learning method that combined recurrent neural networks (RNNs) with CNN for land cover and crop classification. Sonobe et al. [30] calculated 82 published indices and evaluated their importance in crop classification. The results showed that vegetation indices are effective for identifying specific crop types. Dey et al. [31] proposed a daywise and time-series method for crop classification using full-polarimetric synthetic aperture radar (SAR) data. Ok et al. [32] examined the performance of random forest (RF) and maximum likelihood classification (MLC) methods for crop classification, utilizing both pixel-based and parcel-based approaches. Siachalou et al. [33] used hidden Markov models to utilize the phenological information present in temporal image sequences for crop classification. Lv et al. [34] developed a sample iterative enhancement method with good robustness for improving the classification performance of hyperspectral remote sensing images.

D. Summary

Although deep learning methods for crop classification have made significant progress, there are still some limitations, such as the requirements of a large number of training instances, high computational cost, and poor interpretability of the models. Other traditional crop classification methods have not provided satisfactory performance. GP-based methods have shown promise in constructing high-level features for crop classification. However, existing GP-based methods typically extract a single feature type and their potential has not been fully explored. More importantly, existing GP-based crop classification methods may fail to obtain the optimal solution without an effective solution generation and updating strategy. Therefore, this article proposes a new GP approach with a multitree representation and a new population generation method for feature construction in crop classification.

III. Proposed Approach

The overview of the proposed AMTGP approach for crop classification is shown in Fig. 2. This section provides a

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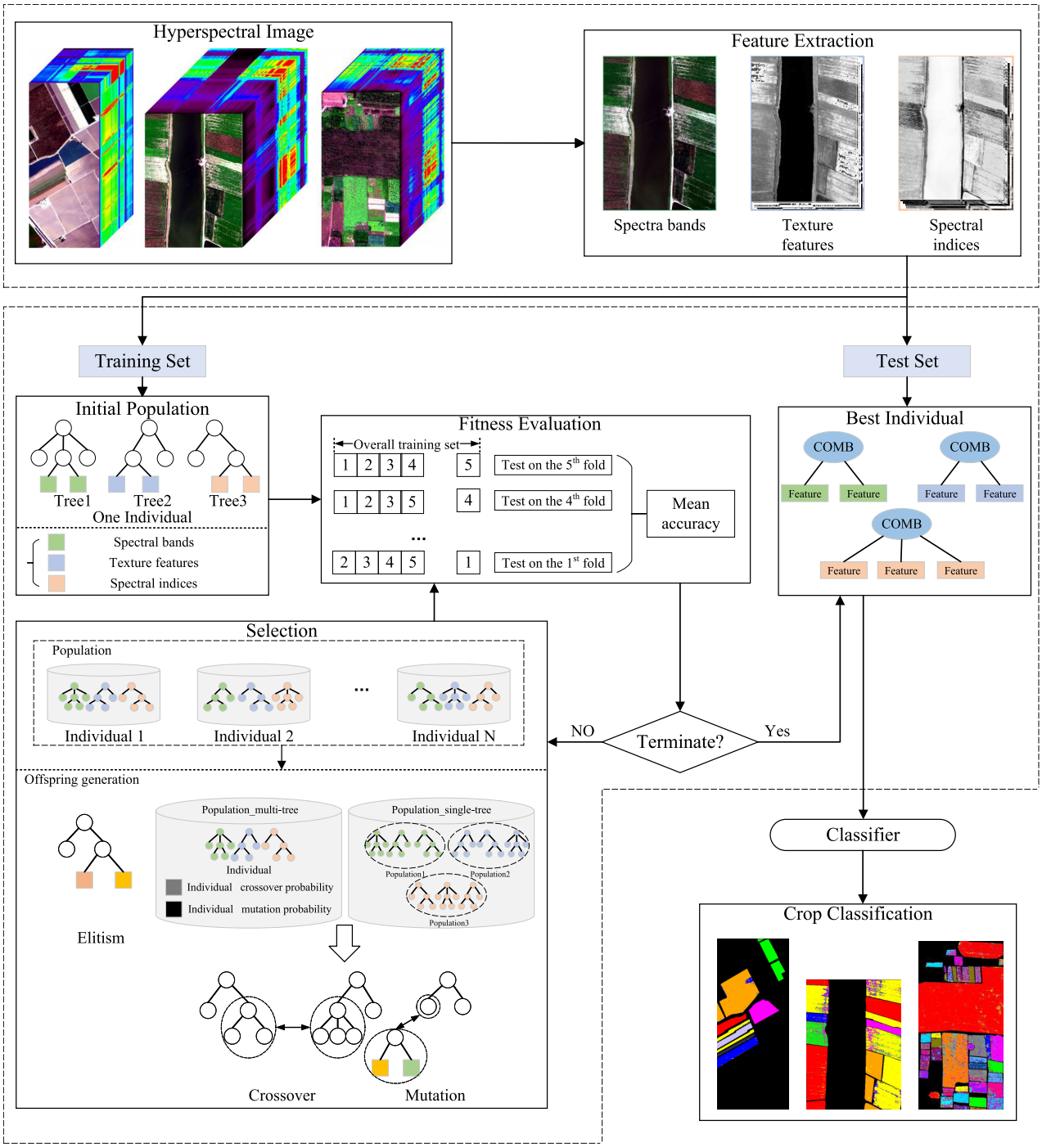


Fig. 2. Overview of GP for feature construction.

detailed description of the proposed approach. The entire process of crop classification is divided into three parts: feature extraction, feature construction, and classification. In the first part, three types of features containing different information are extracted. In the second part, new individual representations and a new offspring generation method are proposed to allow AMTGP to automatically and effectively construct high-level features. In the third part, the constructed high-level features are fed into classical classification algorithms for classification.

A. Feature Extraction

Feature extraction is widely used in the field of image classification, such as remote sensing images, medical images, and facial recognition [35]. Extracting informative features can potentially improve classification accuracy.

Hyperspectral images are composed of numerous spectral bands, which may be effective for crop classification. However, their performance is not satisfactory when dealing with problems involving more categories and complex data structures. Therefore, we have also extracted two additional types of

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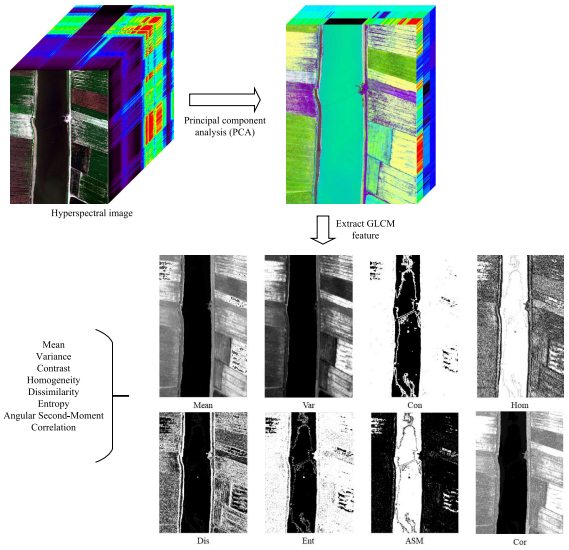


Fig. 3. Process of extracting texture features, using the Longkou dataset as an example.

features to improve the classification accuracy [36]. The three types of features are mainly used to initialize the population of multiple GP trees according to the new individual representation designed in the AMTGP approach. Each tree of GP evolves only one type of feature to ensure its specialization. Based on this, AMTGP can construct three different types of high-level features through a new offspring generation method to improve classification accuracy. The following introduces the three types of features used in the AMTGP approach.

1) Spectral Bands: Each pixel in the hyperspectral image contains abundant spectral information, which is represented by a vector whose size is equal to the number of spectral bands. The spectral bands can reflect the subtle differences between different ground objects in the spectral dimension [37]. Consequently, we extract each band value of the pixel as the first type of feature.

2) Texture Features: Texture features represent important structural information of the spatial distribution between ground objects, which can make up for the deficiency of spectral features in hyperspectral remote sensing image classification. It can describe the details and structures in the image, provide richer information for image classification, and improve the accuracy and robustness of classification. Fig. 3 shows the overall texture feature extraction process from hyperspectral images.

Hyperspectral images consist of a large number of spectral bands. To improve the computational efficiency, dimensionality reduction is performed before extracting texture features. Principal component analysis (PCA) is a commonly used dimensionality reduction technique to transform the high-dimensional data into a smaller set of mutually orthogonal principal components. In hyperspectral images, there may be a certain degree of correlation between different bands. By using PCA, this correlation may be reduced and useful information can be extracted. In the feature extract process, PCA is used on the hyperspectral image to transform the band values into three dimensions, which can preserve rich information while reducing data complexity. For each component, we extract eight types of texture features based on the gray-level co-occurrence matrix (GLCM) that are useful for classifying different crops. These features have general applicability to many types of image data [38], [39]. GLCM is one of the most common spatial and textural feature analysis methods for images and provides information on gray direction, interval, variation amplitude, etc. We extract the eight features according to (1)–(8): mean (Mean), variance (Var), contrast (Con), homogeneity (Hom), dissimilarity (Dis), entropy (Ent), angular second-moment (ASM), and correlation (Cor). These features are widely used in different image classification tasks to achieve good performance

$$\text{Mean} = \frac{1}{MN}\sum_{i=0}^{M-1}\sum_{j=0}^{N-1} f(i,j,d,\theta) \tag{1}$$

$$\text{Var} = \sum_{i=0}^{M-1}\sum_{j=0}^{N-1}(i-\mu)^2 f(i,j,d,\theta) \tag{2}$$

$$\text{Con} = \sum_{i=0}^{M-1}\sum_{j=0}^{N-1}(i-j)^2 f(i,j,d,\theta) \tag{3}$$

$$\text{Hom} = \sum_{i=0}^{M-1}\sum_{j=0}^{N-1}\frac{f(i,j,d,\theta)}{1+(i-j)^2} \tag{4}$$

$$\text{Dis} = \sum_{i=0}^{M-1}\sum_{j=0}^{N-1}|i-j| f(i,j,d,\theta) \tag{5}$$

$$\text{Ent} = -\sum_{i=0}^{M-1}\sum_{j=0}^{N-1} f(i,j)\lg f(i,j,d,\theta) \tag{6}$$

$$\text{ASM} = \sum_{i=0}^{M-1}\sum_{j=0}^{N-1} f(i,j,d,\theta)^2 \tag{7}$$

$$\text{Cor} = \sum_{i=0}^{M-1}\sum_{j=0}^{N-1}\frac{(i-\mu)(j-\mu) f(i,j,d,\theta)^2}{\delta^2} \tag{8}$$

where $f(i,j,d,\theta)$ denotes the normalized grayscale value, i and j represent the grayscale values of the starting and ending pixels when calculating GLCM, respectively, and d and θ are the spatial distance and direction, respectively. M and N represent the grayscale, and δ^2 is the variance value.

3) Spectral Indices: Spectral indices are numerical indicators derived from the measurement and analysis of vegetation reflectance or radiance characteristics. They are calculated based on the relationship between the reflectance or radiance of vegetation in different wavelength bands, which amplify the differences between various types of ground objects. By using spectral indices, precise analysis and monitoring of distribution, changes, and growth status of crops can be conducted, thereby enhancing the accuracy of crop classification [40].

Each spectral index contains diverse information and has its own advantages and limitations [41]. For example, NDVI is calculated from near-infrared bands and infrared bands and can be used to evaluate the growth status of vegetation [42]. Therefore, we use 17 spectral indices as the third type of features, which are sensitive to the leaf area index (LAI), biomass, chlorophyll, carotenoid, anthocyanin, and nitrogen

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TABLE I
SPECTRAL INDICES

Abbreviation	Definition
NDVI [43]	$(R_{800}-R_{670})/(R_{800}+R_{670})$
RDVI [44]	$(R_{800}-R_{670})/\sqrt{R_{800}-R_{670}}$
BGI [45]	R_{400}/R_{550}
ARI [46]	$1/R_{550}-1/R_{700}$
HI [47]	$(R_{534}-R_{698})/(R_{534}+R_{698})-0.5R_{704}$
PSDNc [48]	$(R_{800}-R_{470})/(R_{800}+R_{470})$
PI [49]	$(R_{750}-R_{705})/(R_{750}+R_{705})$
PRI1 [50]	$(R_{570}-R_{531})/(R_{570}+R_{531})$
NRI [51]	$(R_{560}-R_{670})/(R_{560}+R_{670})$
TVI [52]	$0.5\,[(120\,(R_{750}-R_{550})-(R_{670}-R_{550})]$
MCARI [53]	$[(R_{700}-R_{670})-0.2\,(R_{700}-R_{550})]\,(R_{700}/R_{670})$
MCARI2 [54]	$\frac{1.5[2.5(R_{800}-R_{670})-1.3(R_{800}-R_{550})]}{\sqrt{(2R_{800}+1)^2-(6R_{800}-5\sqrt{R_{670}})-0.5}}$
Vog4 [55]	$R_{715}{}'/R_{705}{}'$
R515/R570 [55]	R_{515}/R_{570}
MCARI/OSAVI [56] [57]	$\frac{[(R_{700}-R_{670})-0.2(R_{700}-R_{550})](R_{700}/R_{670})}{[(1+0.16)*(R_{800}-R_{670})]/(R_{800}+R_{670}+0.16)}$
G [58]	R_{544}/R_{670}
PSSRb [58]	R_{800}/R_{650}

content [39]. The formulas of these spectral indices are listed in Table I, where R_n represents the band with a central wavelength of n.

B. Multitree GP Approach to Feature Construction

This section describes the proposed AMTGP approach in detail, including the individual representation, offspring generation method, function set, and terminal set. The flexible multitree representation allows each tree of AMTGP individuals to evolve one type of feature. A potential issue is that it is difficult to find the optimal solution due to the large search space. Furthermore, GP is prone to premature convergence and getting stuck in local optima. To address this, the new offspring generation method dynamically calculates a group of individuals' crossover probability and mutation probability in the evolutionary process to search for better solutions. We ensure the randomness in the composition of individuals of another group to enhance diversity, aiming to avoid getting trapped in local optima.

1) AMTGP Program Representation: The representation of AMTGP is based on strongly typed GP (STGP), whose input and/or output types of functions/terminals must accord with type constraints to construct a flexible number of features. Each individual contains three trees, each of which is constructed with the same function set but different terminal sets. Fig. 4 shows an example of an AMTGP individual, where (a)–(c) represent the three trees of the AMTGP individual. Each tree evolves from a type of feature extracted during the feature extraction phase. Through the function set and terminal set, AMTGP can find various combinations between low-level

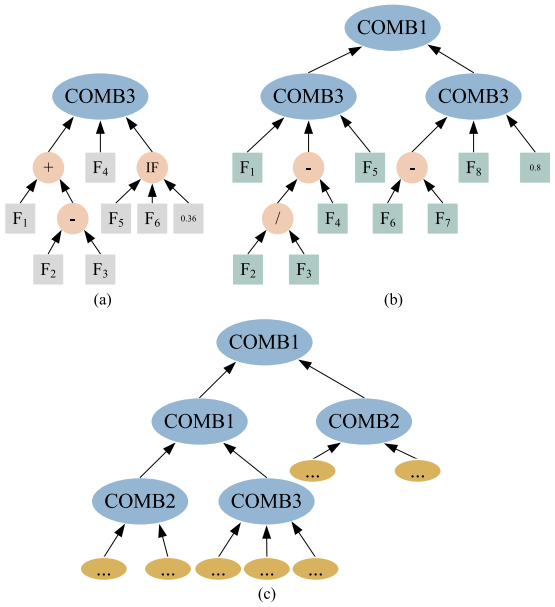


Fig. 4. Representation of an AMTGP individual, including multiple trees. (a)–(c) Three different trees evolving different types of features. The COMB1, COMB2, and COMB3 functions can be used as root or leaf nodes to generate a flexible number of features.

TABLE II
FUNCTION SET

Function	Type	Desciptions
COMB1	Root/Internal	Concatenate two feature vectors into a feature vector
COMB2	Root/Internal	Concatenate two inputs into a feature vector
COMB3	Root/Internal	Concatenate three inputs into a feature vector
Add (+)	Internal	Add of two arguments
Sub (-)	Internal	Substract two arguments
Mul ($\times$)	Internal	Multiply two arguments
ProD (/)	Internal	Divide the second argument using the first argument. Return one if the divisor is zero.
IF	Internal	If the first argument is greater than 0, the second argument is returned. Otherwise, the third argument is returned.

features of each type, and the number of generated features is flexible.

2) Function Set: The functions in the function set comprise the five most commonly used operators $\{+, -, \times,$ protected $/,$ and IF$\}$, which are commonly used in the GP community, and the COMB functions $\{\text{COMB1}, \text{COMB2}, \text{COMB3}\}$ that allow AMTGP to produce a flexible number of high-level features [14]. The division operation ($/$) is protected by returning a value of one when the divisor is equal to zero. The COMB2 and COMB3 functions can appear at the root nodes or internal nodes to concatenate features. The COMB1 function can appear at the root node or internal nodes, and it can connect COMB2 and COMB3 so that AMTGP can generate a flexible number of features. The details of these functions are listed in Table II.

3) Terminal Set: For AMTGP individuals, each tree corresponds to a different terminal set. These three terminal sets are composed of three different types of features extracted from

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hyperspectral images and random constants. Since different datasets have various numbers of bands, the first type of feature has a different number of band values for a pixel. The second and third types of features are 24 texture features and 17 spectral indices extracted from each dataset for classification, respectively. The range of random constants in the terminal set is [0.0, 1.0].

4) Fitness Function: The purpose of fitness evaluation is to assess how well the features constructed by the AMTGP individual are used for crop classification. The fitness function is the classification accuracy obtained by using a classic classification algorithm [including support vector machine (SVM), naive Bayes (NB), and k-nearest neighbor (KNN)] with a k-fold cross-validation manner on the training set.

During the fitness evaluation process, the samples in the training set are transformed into features using three GP trees in an individual, and then, the transformed features from different GP trees are concatenated. Since the features of different types have different scales, we perform the normalization method on these features to prevent bias toward specific types of features, and the min–max normalization is used for its simplicity. Finally, k-fold cross-validation with the transformed training set is performed to evaluate the individual, where k is set to five. Specifically, the transformed training set is divided into fivefold, where fourfold are used to train the classifier, and the remaining fold is used to evaluate the classifier and calculate the classification accuracy. This process is repeated k times until each fold is used once to evaluate the classifier. The mean test accuracy of the fivefold is used as the fitness value. The formula for calculating the classification accuracy of each fold is given as follows:

$$\text{acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{9}$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively.

5) Evolutionary Process and Genetic Operators: During the evolutionary process, the population of each generation is divided into three groups for updating in order to cover a wide range of possible solutions/trees. The purpose of this strategy is to guide the population toward better solutions while ensuring the diversity of the population during the evolutionary process. In the first group, a small set of individuals with better fitness values will be copied to the next generation by using the *elitism* operator. In the second group, tournament selection is used to select a certain number of individuals with better fitness from the current population. Multitree crossover and mutation operations are performed on these individuals to generate new individuals. To make each tree of the AMTGP individual more specialized, the multitree mutation/crossover operators ensure that each tree of the AMTGP individual evolves only one type of feature to enhance the interpretability of the constructed features. To improve the generation of promising trees, we employ dynamic crossover and mutation operators. Specifically, the crossover and mutation probabilities for each selected individual are updated based on the following equations, which encourages mutual learning between the better individuals and mutation of the poorer individuals, thus enhancing diversity [58]:

$$\text{mp}_k = \frac{F_k}{\sum_{i=1}^{I} F_i} \tag{10}$$

$$\text{cp}_{k,k+1} = \frac{2 - \left(\text{mp}_k + \text{mp}_{k+1}\right)}{2(I-1)} \tag{11}$$

where mp represents the mutation probability of AMTGP individuals, cp represents the crossover probability of pairs of individuals, k represents the kth individual in the population, F represents the fitness value of an individual, and I represents the number of individuals in the population.

The third group of individuals ensures randomness in their composition to maintain the diversity of the population during the evolutionary process. In the third group, the three trees of AMTGP individuals in the population are separated to form three subpopulations, where each individual has only one tree. Tournament selection is applied to select a certain number of individuals from each subpopulation for crossover and mutation to generate new individuals. The single-tree GP individuals from the three subpopulations are then merged to form a population containing multitree GP individuals. Algorithm 1 describes the proposed population updating strategy. Algorithms 2 and 3 introduce the adopted mutation/crossover operators.

Algorithm 1 Population Updating

Input : P_g: the current population.
Output : P_{g+1}: the new population.

begin
  $Elit \leftarrow$ **Elitism**(P_g, n);
  $Off \leftarrow$ **TournamentSelection**$(P_g, (|P_g| - n)/2)$;
  $k \leftarrow 1$;
  while $k < (|P_g| - n)/2$ **do**
    $mp_{k-1} \leftarrow$ **MutationProbability**(Off_{k-1});
    $mp_k \leftarrow$ **MutationProbability**(Off_k);
    $cp_{k-1,k} \leftarrow$ **CrossoverProbability**(mp_{k-1}, mp_k);
    $k = k + 2$;
  end
  $Off \leftarrow$ **Mutation/Crossover**(Off);
  $bands_{pop}, texture_{pop}, incices_{pop} \leftarrow$ **Separate**(P_g);
  for *each* sub_{pop} *in* $bands_{pop}, texture_{pop}, incices_{pop}$ **do**
    $sub_{pop} \leftarrow$ **TournamentSelection**$(sub_{pop}, (|P_g| - n)/2)$;
    $sub_{pop} \leftarrow$ **SubTreeMutation/Crossover**(sub_{pop});
  end
  $Off_{random} \leftarrow$ **Merge**$(bands_{pop}, texture_{pop}, incices_{pop})$;
  $P_{g+1} \leftarrow Elit+Off+Off_{random}$;
end
Return P_{g+1}.

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Algorithm 2 Mutation

Input : I: an individual with three trees.
Output : I': the new individual after mutation.

```
begin
    k ← 0;
    while k < 3 do
        s ← Randomly generate a subtree;
        I' ← MUTATION(I_k, s);
        k = k + 1;
    end
end
Return I'.
```

Algorithm 3 Crossover

Input : I^1, I^2: two selected individuals for crossover.
Output : $I^{1'}$, $I^{2'}$: the new individuals after crossover.

```
begin
    k ← 0;
    while k < 3 do
        CROSSOVER(I_k^1, I_k^2);
        k = k + 1;
    end
end
Return I^1', I^2'.
```

C. Classification

The classification process includes the classifier training and testing. The best GP individual is used to transform the original features into a set of high-level constructed features in the training set and the test set. The transformed training set is used to train a classifier, which is then used to classify the samples in the test set. In this article, we have chosen three different classification algorithms for classifier training and testing, i.e., SVM, NB, and KNN. The reason for choosing these three classification algorithms is that they have complementary roles in different situations. SVM performs well in handling high-dimensional datasets and linearly separable problems, while NB has advantages in processing sparse data and training models efficiently. KNN is suitable for handling nonlinear decision boundaries and complex data distributions.

IV. Experimental Design

A. Study Area

The proposed AMTGP approach is evaluated on three hyperspectral datasets, two of which are derived from the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) Academic Research Group, State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University, Wuhan, China [59], [60]. The other dataset was acquired in the Salinas Valley, CA, USA. These datasets contain diverse crop types and have different difficulties, which can be a good choice to evaluate and compare the performance of different

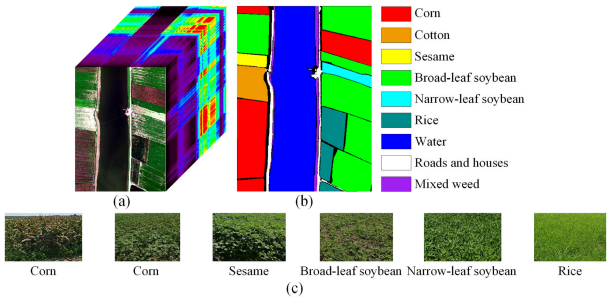


Fig. 5. WHU-Hi-Longkou dataset. (a) Image cube. (b) Ground-truth image. (c) Typical crop photographs in the study area.

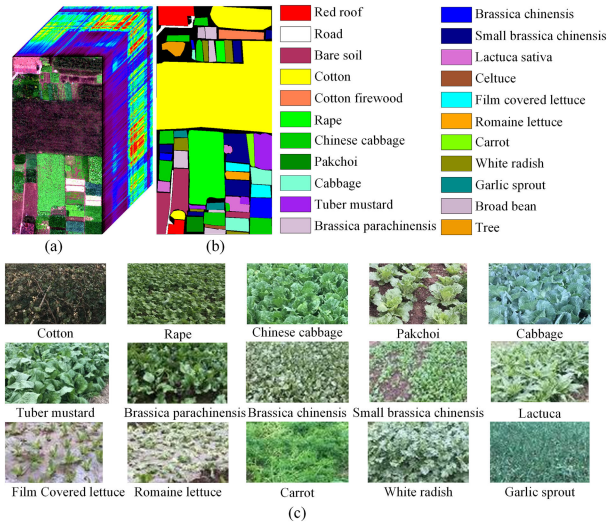


Fig. 6. WHU-Hi-Honghu dataset. (a) Image cube. (b) Ground-truth image. (c) Typical crop photographs in the study area.

classification algorithms. In the experiments, for each crop category, we randomly selected 400 samples and divided them into a 3:7 ratio to form the training set and the test set. Using a larger proportion of the test set can more accurately reflect the model's generalization ability.

1) *Longkou:* With a moderate size, the WHU-Hi-Longkou dataset contains six crop species, which is suitable for evaluating the performance of remote sensing image classification algorithms. The dataset was collected in Longkou Town, Hubei Province, China. The size of the hyperspectral image is 550 × 400 pixels, with 270 bands from 400 to 1000 nm. Fig. 5 provides an overview of the dataset.
2) *Honghu:* The study area is a complex agricultural scene with 15 different crop species scattered throughout. The dataset was collected in Honghu City, Hubei Province. Similar to the WHU-Hi-Longkou dataset, they both have 270 bands from 400 to 1000 nm. Differently, this one has a larger size: 940 × 470 pixels. An overview of the WHU-Hi-Honghu dataset is provided in Fig. 6.
3) *Salinas:* The Salinas dataset is a well-known hyperspectral remote sensing dataset widely used in image classification and land cover identification. It is provided by the National Aeronautics and Space Administration (NASA), Washington, DC., USA, and acquired over

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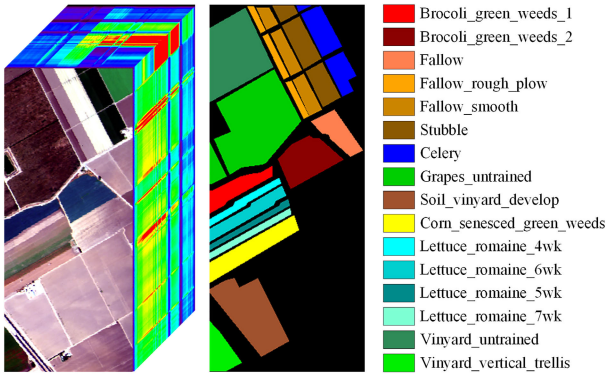


Fig. 7. Salinas AVIRIS dataset. (a) RGB false-color image. (b) Ground-truth image.

Salinas Valley. The dataset has 224 bands and includes 16 different land cover classes, nine of which represent various crops. The size of it is 512 × 217 pixels. Fig. 7 provides an overview of it.

B. Benchmark Methods

In this study, the proposed AMTGP approach is validated for its effectiveness by comparing it with a GP-based method proposed in recent years [14], several GP-based baseline methods, the representative traditional classification algorithms (SVM, NB, and KNN), and the popular deep learning methods [multilayer perceptron (MLP), LSTM, and 1-D CNN (1D-CNN)]. Among them, the GP-based method adopts a flexible representation to construct variable numbers of features. These methods use the same training and test sets as AMTGP to compare the classification accuracy. We embed SVM, NB, and KNN into the GP-based methods to analyze their performance.

SVM is a commonly used supervised learning algorithm that has good generalization abilities. The linear kernel is one of the most fundamental kernel functions in SVM, which can map data to a linear hyperplane in the original feature space for classification. A linear kernel is used in SVM because it has good processing capability for large-scale datasets and relatively low computational cost. KNN is a simple and intuitive classification algorithm, with the number of neighbors set to five. NB is a probability-based classification algorithm and does not require extra parameters. These three classification algorithms are implemented using *scikit-learn* of the Python package with the default parameter settings.

In the experiments, we also compare AMTGP with methods based on deep learning. The reason for choosing MLP as the representation of deep learning methods is its powerful ability to model nonlinearity and adaptability in handling large-scale data. Similarly, we implement MLP using *scikit-learn* of the Python package, following the default settings: three hidden layers each with 512, 256, 128 neurons, rectified linear unit (ReLU) function as the activation function, an Adam optimizer, and a maximum of 200 epochs.

LSTM [61] and 1D-CNN [62] are deep learning models specifically designed to handle sequential data, making them highly suitable for processing time series or any sequentially correlated data. LSTM can capture the correlation between each band and the dependence relationship between different bands of each sample. It is configured with a hidden layer of 128 neurons and two LSTM layers. It uses a cross-entropy loss function and the Adam optimizer, with a training cycle limit of 50. The 1D-CNN is comprised of two convolutional layers, followed by max-pooling layers and two fully connected layers. It uses the ReLU activation function and Adam optimizer for training and the number of training cycles is 50.

TABLE III
PARAMETER SETTING

Population size: 100	Generaitons: 50
Crossover probability: 0.8	Mutation probability: 0.19
Elitism probability: 0.01	Run: 30
Minimum depth: 2	Maximum depth: 10

C. Parameter Settings

For the proposed AMTGP approach and the GP-based method, the parameter settings are the same, using the commonly used settings in the community of GP [16]. To reduce computational cost, the population size is set to 100. The tree generation method is *ramped-half-and-half*, and the selection method is *tournament selection* with a size of 5. Since GP is a stochastic algorithm, each experiment is run independently 30 times to obtain average results. Table III provides the details of the parameter settings.

V. Results and Discussion

In this section, the classification results of the AMTGP approach and the benchmark methods on the three datasets of varying difficulty are analyzed. Additionally, to test the approach's capabilities on simple and more complex tasks, the experiments are to classify four crop classes of each dataset (i.e., corn, cabbage, carrot, and garlic sprout) and all crop classes of each dataset, separately. The Wilcoxon rank-sum test with a 5% significance level is used to show the significance of performance improvement.

A. Classification Results

Tables IV–VI present the results of classification obtained by different methods using the classification methods, i.e., SVM, NB, and KNN, respectively. Among them, "B" represents inputting spectral band features into a classification algorithm for classification. "GP_B" represents the method using spectral band features into a GP-based method for feature construction and then feeding the constructed high-level features into a classification algorithm for classification.

1) Comparisons With GP-Based Methods: In the proposed approach, the three trees of each GP individual can construct features from three types of original ones. To better compare with the GP-based method [14] and demonstrate the effectiveness of the multitree feature construction stage, we use the combinations of different types of features as inputs for the GP-based methods with a single-tree representation, which

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TABLE IV

CLASSIFICATION ACCURACIES OF METHODS USING SVM FOR CLASSIFICATION. "B," "T," AND "I" DENOTE THE INPUT AS SPECTRAL BANDS, TEXTURE FEATURES, AND SPECTRAL INDICES, RESPECTIVELY. "+," "−," AND "=" DENOTE THAT AMTGP ACHIEVES BETTER, WORSE, AND SIMILAR MEAN ACCURACY THAN/TO THE COMPARED METHODS

	Salinas		Longkou		Honghu	
	4	all	4	all	4	all
B	45.19±1.52+	92.63±5.32+	88.10±3.97+	87.71±4.01+	95.49±0.14–	60.40±2.87+
T	94.06±6.20+	65.93±11.76+	75.18±5.85+	67.94±6.31+	74.31±8.88+	18.65±5.46+
I	47.21±17.15+	37.34±14.71+	56.76±10.47+	51.53±9.97+	77.80±8.35+	26.24±5.67+
(B,T)	85.18±12.97+	95.13±0.76+	76.08±6.20+	72.39±5.90+	66.10±12.14+	18.79±5.05+
(B,I)	47.07±4.42+	69.97±11.29+	74.05±7.76+	70.78±7.24+	90.75±3.37+	42.98±9.29+
(T,I)	66.35±14.23+	68.73±15.18+	74.79±5.39+	66.21±10.15+	72.73±9.55+	19.33±5.68+
(B,T,I)	92.34±8.68+	83.06±13.04+	77.59±4.56+	69.99±7.30+	67.94±11.95+	20.17±5.53+
GP_B	96.05±7.23+	94.70±3.81+	89.34±2.65+	89.31±2.00+	87.74±2.41+	62.54±1.33+
GP_T	94.43±6.53+	88.24±5.69+	84.41±4.72+	78.33±6.43+	80.47±6.50+	40.90±4.66+
GP_I	97.97±0.31+	93.80±0.67+	81.57±1.05+	81.58±0.88+	81.57±1.05+	58.35±0.90+
$GP_{B,T}$	94.89±11.76+	93.42±5.27+	87.82±2.65+	88.57±2.26+	94.56±0.66+	67.86±1.30+
$GP_{B,I}$	97.31±0.89+	88.81±10.02+	89.30±2.51+	88.62±2.64+	87.45±2.70+	64.04±0.87+
$GP_{T,I}$	95.73±5.67+	92.56±5.96+	85.02±2.83+	83.64±3.25+	91.22±1.77+	59.43±4.58+
$GP_{B,T,I}$	93.00±14.09+	93.63±3.04+	88.16±1.81+	87.20±3.54+	94.25±0.56+	68.39±1.30+
$2TGP_{B,T}$	99.27±0.62=	96.35±0.39+	91.00±1.16=	93.19±1.19=	95.41±0.46=	74.00±0.77=
$2TGP_{B,I}$	97.90±0.50+	97.91±0.49–	93.35±1.15–	91.29±0.98+	85.79±1.05+	65.53±1.52+
$2TGP_{T,I}$	98.38±0.45+	96.09±0.30+	90.60±0.69=	91.00±0.67+	93.56±0.60+	67.75±0.94+
$AMTGP_{B,T,I}$	99.24±0.40	96.86±0.26	90.60±1.02	93.13±1.14	95.24±0.45	73.54±1.62
Overall	16/1/0	16/0/1	14/2/1	16/1/0	15/1/1	16/1/0

TABLE V

CLASSIFICATION ACCURACIES OF METHODS USING NB FOR CLASSIFICATION. "B," "T," AND "I" DENOTE THE INPUT AS SPECTRAL BANDS, TEXTURE FEATURES, AND SPECTRAL INDICES, RESPECTIVELY. "+," "−," AND "=" DENOTE THAT AMTGP ACHIEVES BETTER, WORSE, AND SIMILAR MEAN ACCURACY THAN/TO THE COMPARED METHODS

	Salinas		Longkou		Honghu	
	4	all	4	all	4	all
B	91.07±0.00+	88.85±0.00+	76.07±0.00+	62.92±0.00+	79.82±0.00+	56.05±0.00+
T	97.41±0.00+	93.61±0.00+	78.04±0.00+	74.35±0.00+	77.68±0.00+	45.45±0.00+
I	95.98±0.00+	88.97±0.00+	68.93±0.00+	62.20±0.00+	73.12±0.00+	30.64±0.00+
(B,T)	90.80±0.00+	89.80±0.00+	77.32±0.00+	65.89±0.00+	68.57±0.00+	38.67±0.00+
(B,I)	90.62±0.00+	89.13±0.00+	76.34±0.00+	64.64±0.00+	80.18±0.00+	45.90±0.00+
(T,I)	96.96±0.00+	93.97±0.00+	77.86±0.00+	75.06±0.00+	77.77±0.00+	44.55±0.00+
(B,T,I)	91.34±0.00+	90.24±0.00+	76.79±0.00+	66.25±0.00+	68.39±0.00+	38.45±0.00+
GP_B	97.66±0.48=	96.21±0.75=	88.95±1.54–	80.18±2.10+	82.75±0.81+	54.34±2.61+
GP_T	97.10±1.30=	95.36±0.48+	86.87±1.51+	80.07±1.49+	85.22±1.46+	48.98±1.10+
GP_I	98.44±0.44=	94.14±1.05+	79.94±1.71+	77.24±2.06+	79.94±1.71+	47.63±1.25+
$GP_{B,T}$	98.82±1.09–	95.11±0.68+	85.79±2.44+	81.28±1.60+	87.48±2.50+	60.45±1.65+
$GP_{B,I}$	97.87±0.39=	96.30±0.83–	89.31±1.43–	79.33±2.50+	82.65±0.96+	56.83±1.69+
$GP_{T,I}$	98.20±1.32–	95.37±0.59+	85.37±2.22+	81.07±1.72+	87.11±1.31+	56.80±0.66+
$GP_{B,T,I}$	98.76±0.77–	95.19±0.57+	86.59±1.94+	81.35±1.56+	88.40±1.48=	60.16±1.83+
$2TGP_{B,T}$	97.81±1.18=	95.76±0.51+	88.48±1.49=	84.01±1.47=	88.75±1.18=	61.64±1.25=
$2TGP_{B,I}$	98.11±0.37=	97.03±0.53–	90.24±1.41–	81.23±1.51+	82.85±0.54+	55.86±1.82+
$2TGP_{T,I}$	97.73±0.84=	96.01±0.41=	87.23±1.67+	83.34±1.81+	88.43±1.01+	58.58±1.16+
$AMTGP_{B,T,I}$	97.64±1.58	96.11±0.41	88.22±1.78	83.48±1.14	89.09±1.25	61.72±1.19
Overall	7/7/3	13/2/2	13/1/3	16/1/0	15/2/0	16/1/0

are GP_B, GP_T, GP_I, and $GP_{B,\,T,\,I}$. Additionally, we compare AMTGP with the GP-based method using a two-tree representation and two types of features (including all combinations of two types of features) to show the effectiveness of the feature construction from three types of features. The methods are termed as $2TGP_{B,T}$, $2TGP_{B,I}$, and $2TGP_{T,I}$.

In the experiments using SVM for classification, AMTGP achieves significantly better performance in 41 out of the total 48 comparisons with the GP-based methods. In the experiments using NB, AMTGP achieves significantly better performance in 29 out of the total 48 comparisons with the GP-based methods. In the experiments using KNN, AMTGP achieves significantly better performance in 36 out of the total 48 comparisons with the GP-based methods. The results show that the proposed AMTGP approach achieves significantly better performance than the GP-based methods, including the methods using one type of features, two types of features, and the single-tree GP method with three types of features. The results show that the proposed AMTGP approach with three-tree representation can construct meaningful features for effective classification. By comparing the performance of different GP methods using different types of features, we can find that the proposed approach using three types of features is the most effective one for feature construction in crop classification. In addition, the proposed approach achieves better performance than the two-tree GP methods, which shows that the three types of features are useful for feature construction. Overall, the proposed approach is better than the GP-based methods for feature construction in crop classification.

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TABLE VI

CLASSIFICATION ACCURACIES OF METHODS USING KNN FOR CLASSIFICATION. "B," "T," AND "I" DENOTE THE INPUT AS SPECTRAL BANDS, TEXTURE FEATURES, AND SPECTRAL INDICES, RESPECTIVELY. "+," "−," AND "=" DENOTE THAT AMTGP ACHIEVES BETTER, WORSE, AND SIMILAR MEAN ACCURACY THAN/TO THE COMPARED METHODS

	Salinas		Longkou		Honghu	
	4	all	4	all	4	all
B	98.93±0.00+	95.12±0.00+	77.77±0.00+	73.99±0.00+	80.00±0.00+	56.05±0.00+
T	98.48±0.00+	96.11±0.00+	88.12±0.00+	85.77±0.00+	84.38±0.00+	45.45±0.00+
I	96.52±0.00+	84.29±0.00+	52.41±0.00+	49.40±0.00+	60.00±0.00+	30.64±0.00+
(B,T)	98.39±0.00+	95.79±0.00+	88.12±0.00+	85.77±0.00+	84.38±0.00+	45.45±0.00+
(B,I)	97.86±0.00+	90.60±0.00+	68.21±0.00+	64.64±0.00+	68.30±0.00+	43.52±0.00+
(T,I)	97.86±0.00+	92.98±0.00+	88.12±0.00+	85.77±0.00+	84.38±0.00+	45.45±0.00+
(B,T,I)	97.59±0.00+	93.41±0.00+	88.12±0.00+	85.83±0.00+	84.38±0.00+	45.40±0.00+
GP_B	97.88±0.44+	96.16±0.54+	89.32±1.27=	81.60±2.49+	83.65±1.05+	63.76±1.02=
GP_T	98.03±1.06+	96.00±0.40+	87.74±0.40+	86.00±0.54+	83.57±0.83+	50.21±0.94+
GP_I	98.60±0.45+	94.62±0.85+	79.27±1.59+	80.51±2.30+	79.27±1.59+	55.84±1.00+
$GP_{B,T}$	99.09±0.63+	95.66±0.53+	84.03±2.48+	81.79±2.67+	85.68±2.32+	60.30±1.41+
$GP_{B,I}$	97.73±0.45+	96.09±0.44+	89.76±1.81=	81.45±2.53+	83.21±0.92+	64.04±0.87=
$GP_{T,I}$	98.40±0.74+	95.99±0.51+	87.52±1.14=	86.22±0.61+	84.39±2.08+	53.01±2.40=
$GP_{B,T,I}$	98.73±1.15+	95.82±0.45+	83.71±2.22+	81.90±2.48+	86.86±2.71+	60.81±1.14+
$2TGP_{B,T}$	99.05±1.07=	96.58±0.54=	88.93±1.59=	89.59±1.36=	87.28±1.60+	64.53±0.69=
$2TGP_{B,I}$	97.98±0.56+	97.87±0.29−	90.96±1.37−	85.46±2.04+	84.38±0.53+	66.83±0.55−
$2TGP_{T,I}$	97.66±1.17+	95.92±0.71+	88.18±0.81+	88.61±1.02+	89.40±1.33=	61.78±0.81+
$AMTGP_{B,T,I}$	99.15±0.90	96.76±0.37	89.01±1.28	90.21±1.00	89.15±1.19	64.24±0.86
Overall	16/1/0	15/1/1	13/3/1	16/1/0	16/1/0	13/3/1

TABLE VII

AVERAGE CLASSIFICATION ACCURACIES OF MLP, 1D-CNN, LSTM, AND AMTGP WITH DIFFERENT CLASSIFIERS ON THREE DATASETS

		Salinas		Longkou		Honghu	
		4	all	4	all	4	all
AMTGP	SVM	**99.24**	**96.86**	**90.60**	**93.13**	**95.24**	**73.54**
	NB	97.64	96.11	88.22	83.48	89.09	61.72
	KNN	99.15	96.76	89.01	90.21	89.15	64.24
MLP		92.51	85.05	84.46	81.17	77.38	39.28
1D-CNN		98.53	95.16	87.96	86.41	86.43	55.39
LSTM		98.26	93.53	88.99	88.18	86.08	56.58

2) Comparisons With Classic Methods: The classic methods represent the methods using the original bands or extracted features for classification, which are B, T, I, (B, T), (B, I), (T, I), and (B, T, I) in Tables IV–VI. Compared with the methods using SVM, AMTGP achieves significantly better performance in 35 out of the total 36 cases. Compared with the methods using NB and KNN, AMTGP achieves significantly better performance in all 36 comparisons. The results show that the features constructed by AMTGP are more effective for crop classification. Compared with these methods, the proposed AMTGP approach can select features from the original feature implicitly and construct high-level features, which improves the classification performance. The analysis confirms the effectiveness of the high-level features obtained in the feature construction stage.

3) Comparisons With Deep Learning Methods: The proposed approach is compared with deep learning methods, including MLP, LSTM, and 1D-CNN. The classification results are presented in Table VII. The best results in each comparison are highlighted in bold. Compared with these methods, AMTGP with the SVM classifier achieves the best classification performance among all these methods on the six different tasks. AMTGP with the NB classifier achieves better performance than the three methods on three tasks, and AMTGP with KNN achieves better results on six tasks. The results show that AMTGP is more effective than these three methods for crop classification. One possible reason is that the three methods may require a large number of training data, but the GP-based method can learn effective models from limited data. The results demonstrate that AMTGP exhibits better global search capability and can adapt to different crop classification tasks.

B. Classification Results of the Whole Area

To further analyze the performance of the proposed approach and clearly compare the differences between different classification algorithms, we apply the trained classifiers to classify all the pixel points in the images into four crop types and all crop types, respectively. Fig. 8 demonstrates the results of classifying four types of crops and all crops on the Longkou dataset. Each color represents one class. The results clearly show that the proposed AMTGP approach achieves better performance than these compared methods on the two tasks. From the region marked by circles in Fig. 8, it can be observed that when the classification is expanded from four categories to six categories, the GP-based methods show a decrease in classification performance in that region, misclassifying the category into more other categories. However, AMTGP demonstrates an improvement in classification performance in that region. This indicates that AMTGP can construct effective features with better discriminative ability.

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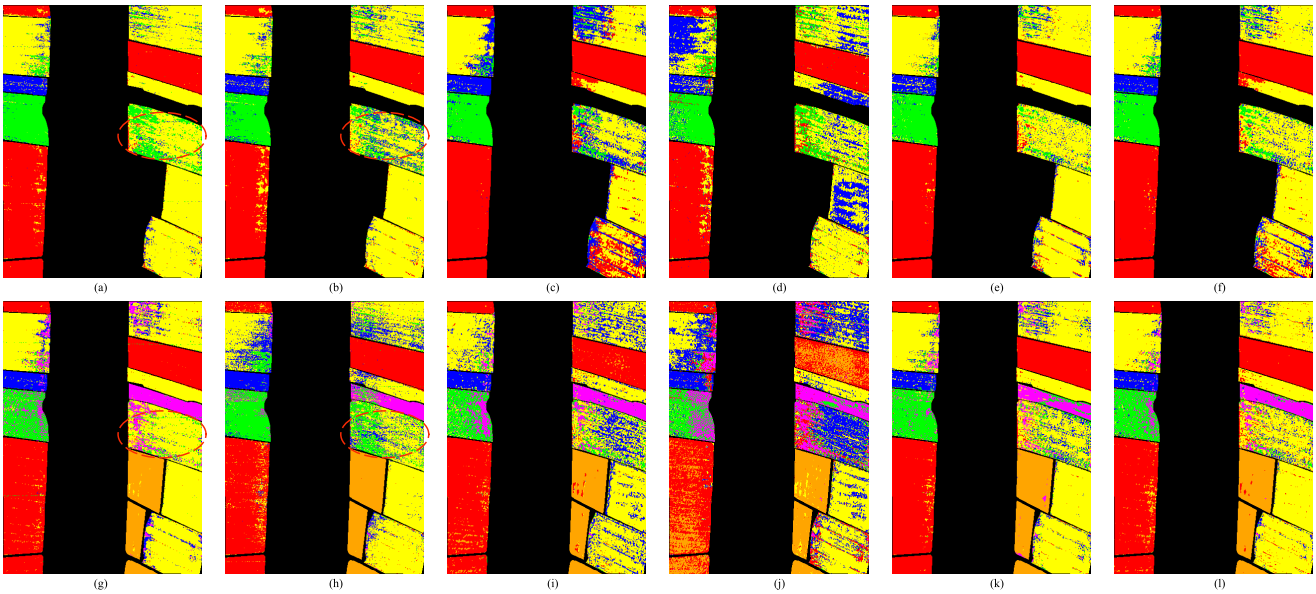


Fig. 8. Classification results of the Longkou dataset with (a)–(f) four-class classification and (g)–(l) six-class classification using different classification algorithms. A circle region has been marked in the figure to compare AMTGP and the GP-based method.

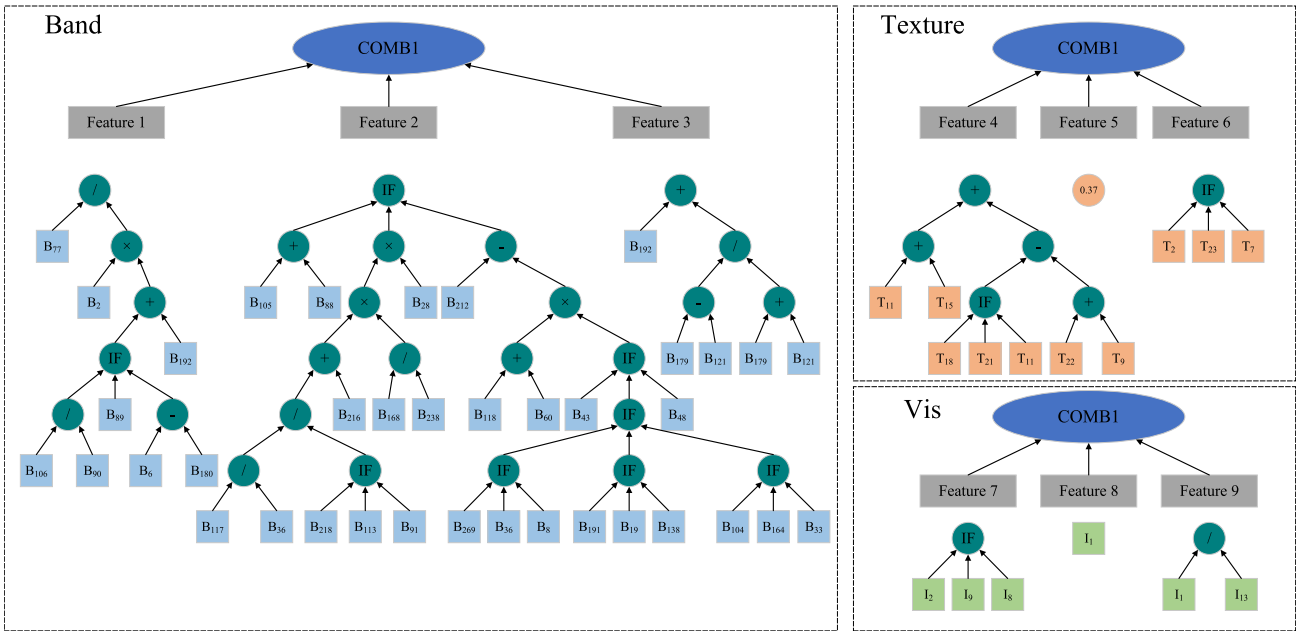


Fig. 9. Example of features constructed by AMTGP.

C. Summary

In summary, AMTGP outperforms the GP-based methods in the majority of the classification tasks on the three datasets. Additionally, the results indicate that the use of three types of features for crop classification is effective in improving the performance of different crop classification tasks. The use of these features, along with the multitree representation and offspring generation method, enhances the effectiveness of AMTGP in solving crop classification problems. Furthermore, compared to classic classification algorithms and deep learning methods, AMTGP exhibits superior performance. These results demonstrate that AMTGP is an effective approach to feature construction for crop classification.

VI. Further Analysis

This section further analyzes the AMTGP approach to better understand why its constructed high-level features can effectively classify crops.

A. Visualization of GP Trees and Constructed Features

Taking the Salinas dataset as an example, we visualize the high-level features constructed by AMTGP. The AMTGP individual, consisting of three trees, is built with spectral bands, texture features, and spectral indices as inputs and NB as the classifier, as shown in Fig. 9. This AMTGP individual constructs nine high-level features, achieving a classification accuracy of 88.39% using NB. The first tree selects 32 bands

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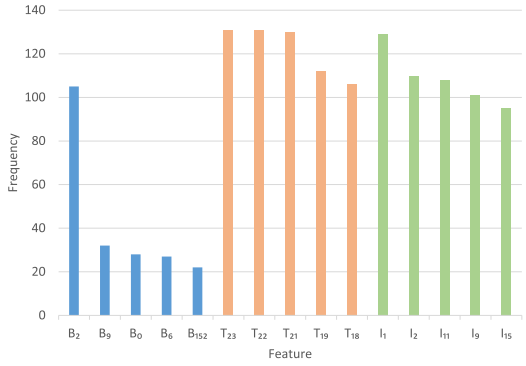


Fig. 10. Frequency of the top 15 features selected by AMTGP for feature construction.

TABLE VIII
RUNNING TIME (min)

		Salinas		Longkou		Honghu	
		4	all	4	all	4	all
SVM	AMTGP	826.0	9557.8	2662.7	5378.4	2061.2	35023.4
	GP	295.6	2523.9	436.4	1684.2	477.9	3128.4
NB	AMTGP	346.4	908.2	342.8	658.7	470.3	1811.8
	GP	85.5	257.5	104.7	253.8	164.1	822.1
KNN	AMTGP	662.7	1707.2	942.1	1438.5	1006.6	3441.3
	GP	193.1	660.7	237.3	384.0	319.6	1186.9

out of 270 bands and uses functions to construct three high-level features. The second tree selects nine features out of the 24 texture features to combine and obtain three features. Feature 5 is a random constant. The third tree selects five indices out of the 17 spectral indices to combine and obtain three features. Feature 8 is the spectral indices PSDNc. The subtree of Feature 3 forms the NDVI spectral index, which can reflect the vegetation growth status. Compared to deep learning methods, we can observe which original features are selected by the AMTGP approach for construction. Moreover, the constructed features can be easily visualized in Fig. 9, which provides insights into how these features are constructed. This demonstrates that the solutions evolved by AMTGP have potentially good interpretability.

In Fig. 10, we analyze the top five features with the highest frequency in each tree of the solutions obtained by the AMTGP approach. Taking the first tree of the AMTGP individual as an example, the most frequently used band is B_2. In this band, the chlorophyll content and photosynthetic activity of crops can be observed, and these features can to some extent differentiate different crops, serving as effective indicators for classifying crops. The analysis indicated that GP has implicit feature selection capability and can automatically identify useful features to construct high-level features.

B. Convergence Behaviors

This section compares the convergence behavior of AMTGP and the GP-based method using all three types of features [14]. Taking the four-class classification task of each dataset as an example, the convergence behaviors of the two methods are shown in Fig. 11. This is a comparative analysis conducted by combining the two methods separately with each classifier on three datasets. There are differences in the initial accuracy between the two methods due to variations in individual representation. It can be seen that AMTGP has much better starting points and convergence ability in most cases, and can ultimately converge to a higher point. Especially on the Longkou dataset, when the two methods are combined with NB for classification, AMTGP still does not converge to the optimal value, indicating its potential for achieving higher classification accuracy.

When the two methods are combined with NB for classification on the Honghu dataset, the GP-based method demonstrates a faster convergence speed, proving that this method also has good convergence ability. However, in all other tasks, AMTGP has shown better convergence ability than the GP-based methods.

C. Running Time

The average running time (including training time and test time) of the 30 runs for AMTGP and the GP-based method [14] is compared in Table VIII. Compared to the GP-based method, AMTGP consists of three trees, and its offspring generation method is more complex. The evolved solution is also more complex, which increases the training time of GP.

D. Effectiveness of New Components of AMTGP

1) Effectiveness of the New Representation: Table IX shows the comparison of the classification results of AMTGP and the GP method with a single-tree representation. The AMTGP approach achieves similar or significantly better performance to/than the compared methods in 16 comparisons out of the 18 comparisons. The results show that the multitree representation can improve the performance of AMTGP for crop classification. The benefit of using a multitree representation includes better exploring the interactions between the single type of features and a smaller search space.

2) Effectiveness of Using the New Offspring Generation Method: Table X presents the results of AMTGP and the GP method without the new offspring generation method [i.e., random multitree GP (RMTGP)]. In RMTGP, the offspring generation method follows the standard one. The results show that AMTGP achieves significantly better or similar results than/to RMTGP in 15 out of the 18 cases. In AMTGP, the offspring generation method generates new offspring from two main groups. One group uses dynamic crossover and mutation operators to facilitate mutual learning among better individuals and mutation among relatively weaker individuals. Another group evolves trees separately to introduce randomness to the population, which further enhances the diversity of the population. Overall, the new offspring generation method can improve the search ability of AMTGP, leading to a good performance in crop classification.

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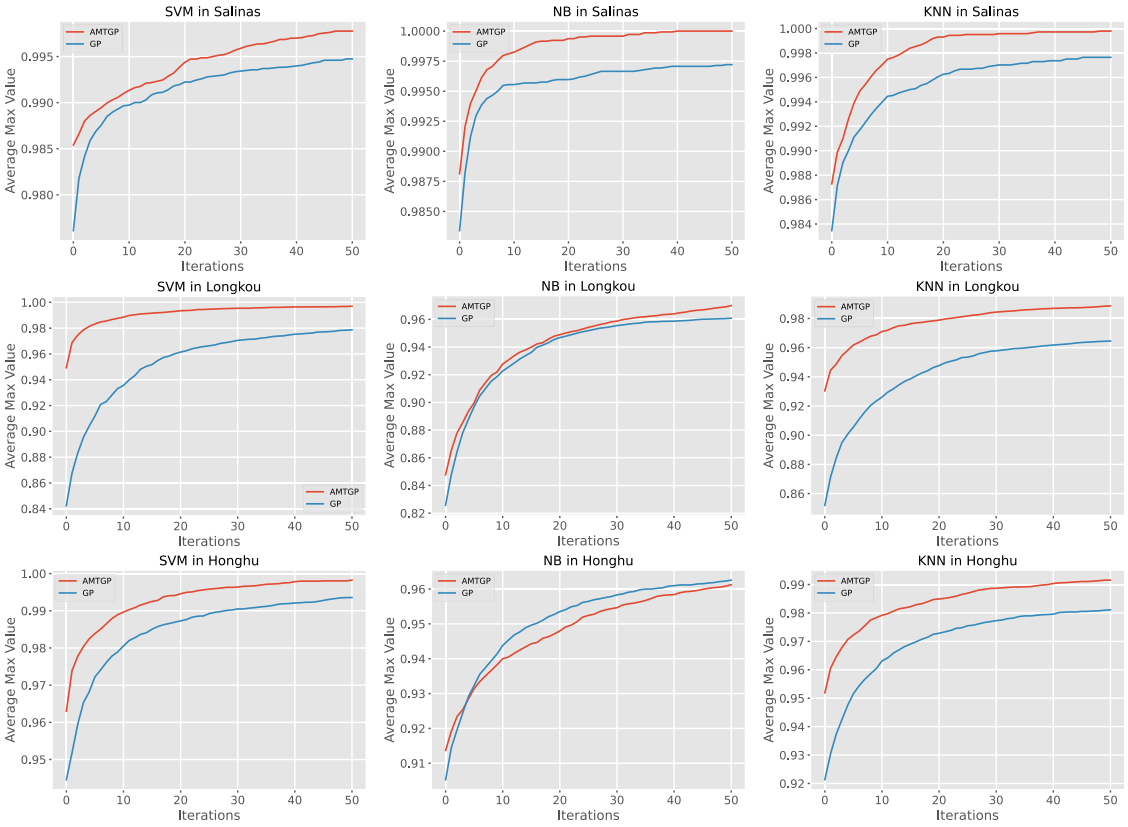


Fig. 11. Comparisons of convergence curves between the two GP methods.

TABLE IX
CLASSIFICATION RESULTS OF AMTGP AND THE GP METHOD WITH A SINGLE-TREE REPRESENTATION (GP)

		Salinas		Longkou		Honghu	
		4	all	4	all	4	all
RMTGP	SVM	98.52±0.61	96.32±0.46	89.66±1.22	90.19±1.46	94.25±0.82	72.99±1.05
	NB	98.62±0.93	95.65±0.61	88.10±2.20	82.33±1.86	88.50±1.41	60.41±1.41
	KNN	98.32±1.45	96.51±0.61	88.66±2.06	88.34±0.99	90.24±1.01	64.95±1.14
GP	SVM	96.05±7.23+	94.70±3.81=	89.34±2.65=	89.31±2.00=	87.74±2.41+	62.54±1.33+
	NB	97.66±0.48+	96.21±0.75–	88.95±1.54=	80.18±2.10+	82.75±0.81+	54.34±2.61+
	KNN	97.88±0.44+	96.16±0.54+	89.32±1.27=	81.60±2.49+	83.65±1.05+	63.76±1.02+
Overall		12/5/1					

TABLE X
CLASSIFICATION RESULTS OF AMTGP AND THE GP METHOD WITHOUT THE NEW OFFSPRING GENERATION METHOD (RMTGP)

		Salinas		Longkou		Honghu	
		4	all	4	all	4	all
AMTGP	SVM	99.24±0.40	96.86±0.26	90.60±1.02	93.13±1.14	95.24±0.45	73.54±1.62
	NB	97.64±1.58	96.11±0.41	88.22±1.78	83.48±1.14	89.09±1.25	61.72±1.19
	KNN	99.15±0.90	96.76±0.37	89.01±1.28	90.21±1.00	89.15±1.19	64.24±0.86
RMTGP	SVM	98.52±0.61+	96.32±0.46+	89.66±1.22+	90.19±1.46+	94.25±0.82+	72.99±1.05+
	NB	98.62±0.93–	95.65±0.61+	88.10±2.20=	82.33±1.86+	88.50±1.41=	60.41±1.41+
	KNN	98.32±1.45+	96.51±0.61=	88.66±2.06=	88.34±0.99+	90.24±1.01–	64.95±1.14–
Overall		11/4/3					

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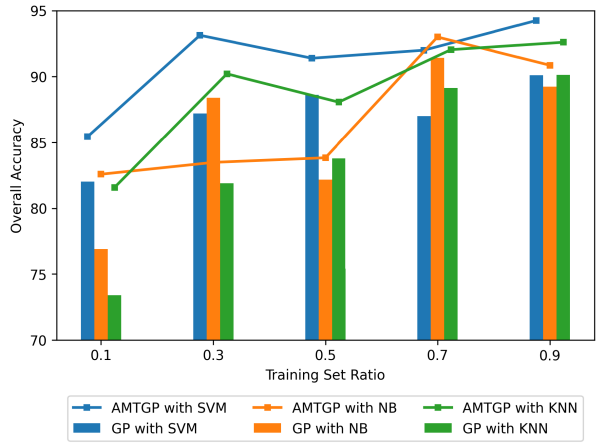


Fig. 12. Accuracies of AMTGP and the GP-based method wrapped three classification algorithms (i.e., SVM, NB, and KNN) under different training set ratios.

E. Training Set Size

This section analyzes the classification results of six crops in the Longkou dataset using AMTGP and the GP-based method [14] under different training set ratios. From Fig. 12, it can be observed that AMTGP achieves better results than the compared methods under different training set ratios. Even when the training set is small, AMTGP demonstrates good performance, indicating that it is minimally affected by the number of training samples.

F. Summary

To sum up, AMTGP achieves better performance in most comparisons on datasets of different difficulties and tasks. The new terminal set, individual representation, and offspring generation method enhance the capability of AMTGP in remote sensing image classification. The results indicate that AMTGP is a highly effective approach to crop classification although there is substantial room for further development in AMTGP. The number of features learned by AMTGP is flexible, and it is possible to construct a large number of features, which causes the model to fall into local optimality. Furthermore, additional research is required to enhance the interpretability of the model in such scenarios.

VII. Conclusion

This article proposed a new GP approach to construct high-level features for crop classification on remote sensing hyperspectral images. Three types of features with different information were extracted from the hyperspectral images. The proposed AMTGP approach can automatically construct effective high-level features by taking these features as inputs. AMTGP adopted a multitree representation, where each tree evolves features of a single type to ensure its specialization. During the evolutionary process, a new offspring generation method was introduced to divide the population into different groups. Dynamic crossover and mutation operators were used in one group of individuals to facilitate mutual learning among better individuals and mutation among relatively weaker individuals. The other group introduced the randomness in the composition of individuals of trees generated independently to enhance the diversity of the population.

The features constructed by AMTGP were used for crop classification using three classifiers: SVM, NB, and KNN. AMTGP was compared with GP-based methods, classic classifiers, and deep learning methods on three datasets with different difficulties and tasks. AMTGP achieved better results than these methods in most cases. Further analysis indicated that AMTGP had a better convergence ability and achieved good performance when the size of the training set is small. The analysis and visualization of the features constructed by AMTGP demonstrated its good interpretability. The results showed the potential of AMTGP in crop classification.

However, the proposed AMTGP approach is computationally more expensive than the GP-based method. When the classification task is difficult, the AMTGP approach may generate a relatively large number of features to achieve good performance. In future research, we will explore methods to reduce its computational cost and construct fewer features on complex tasks without compromising its performance.

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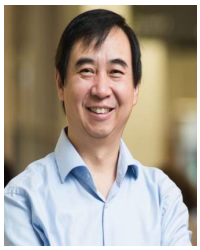
He is the Vice Secretary-General of the Youth Committee at the Chinese Society for Agricultural Machinery.



Bing Xue (Fellow, IEEE) is currently a Professor of artificial intelligence, the Deputy Head of the School of Engineering and Computer Science, and the Deputy Director of the Center for Data Science and Artificial Intelligence, Victoria University of Wellington (VUW), Wellington, New Zealand.

She has over 400 articles published in fully refereed international journals and conferences.

Dr. Xue has also served as an Associate Editor for several international journals, such as IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION (TEVC), IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE (TAI), *IEEE Computational Intelligence Magazine* (CIM), and *ACM Transactions on Evolutionary Learning and Optimization* (TELO). She is a fellow of Engineering New Zealand.



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He has authored over 800 research articles in refereed international journals and conferences.

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